

PV-CTL control board

Rev A2 | 2026-10-06 | Schematics authored by Chinmoy Bhuyan | Cost-first PV-P75/100/110 controller: F280039C on BUS-, one trip latch, one SELV barrier

01 PC connector, supply, reset

2x32 PC contract to the power board (all LIVE): only +3V3 is used (200 mA allocated by PV-PWR); TPS3828 reset + 1.6 s watchdog

File: sheets/01_supply.kicad_sch

02 F280039C, clock, debug, recorder flash

TMS320F280039C (100-pin PZ), decoupling, 25 MHz SPX0, cTI-20 header in cJTAG mode (LIVE), 4 MB recorder flash, boot straps, command pull-downs

File: sheets/02_mcu.kicad_sch

03 Reference and analog front ends

REF3030E 3.0 V = VREFHI = VDAC = NTC bias = trip thresholds; IL1-4 gain stages; power-board signals into the ADC through 100 R / 1 nF; 8 NTC biases + inlet NTC

File: sheets/03_afe.kicad_sch

04 Hardware trip comparators

7 x TLV9024 on +3V3 (open drain, no firmware): IL windows from each sensor's Uref; VA / VB over-voltage, NTC over-temperature and open probe from VREF ladders

File: sheets/04_trip.kicad_sch

05 Trip latch, watchdog, gating

74LVC07 wired-AND of every trip source -> PRE_N -> 74LVC1G74 latch (no firmware); heartbeat monoflop; 74LVC08 gating of 16 PWM, EN, K_A, K_B, K_PRE, STATUS, FAN

File: sheets/05_latch.kicad_sch

06 Barrier B1: isolators, CAN, RS-485

Reinforced barrier B1: CA-IS3050W CAN, CA-IS3082WNX RS-485, 2 x CA-IS3821LG, CA-IS3820LG, CA-IS3842HW (Ethernet UART, DI1); bus TVS, jumper terminations (SELV)

File: sheets/06_iso.kicad_sch

07 SELV supply, fans, stop, DI1, relay

SELV 24 V input (AUX-T1 reinforced winding, 12-36 V), TPS54360B 5 V and fan supply, 3 fan headers + tach, stop (chainable) / ready / DI1 terminals, relay, SELV 0 V to PE

File: sheets/07_selv.kicad_sch

08 Ethernet bridge (EMS, Modbus TCP)

WCH CH9121T UART-to-Ethernet bridge on the SELV side, HR913550AE RJ45 with magnetics, SRV05-4, TPS7A2033 3.3 V. NOT FITTED on PV-CTL / PV-CTL-P75: footprints only (DNP).

File: sheets/08_eth.kicad_sch

ARCHITECTURE-COSTFIRST sections 6.1 (trip sources -> one latch), 10 (two zones); PC levels as built on PV-PWR (hardware/PV-PWR/outputs/PV-PWR_design_check.txt).
Assembly option: the two TLV9024 marked 'PV-P100/110 only' are not fitted on PV-P75 (bom/PV-CTL-P75_BOM.csv).
Assembly option: the Ethernet bridge (sheet 08_eth) is NOT FITTED on PV-CTL / PV-CTL-P75 (DNP); fitted on PCS-CTL. Debug header in cJTAG mode only.
Pin plan: gen/data/pv_ctr_pin_plan.csv (from SPRSP61C, checked every build).

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 1
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /
File: PV-CTL.kicad_sch

Title: PV-CTL control board – Cover / Hierarchy

Size: A3

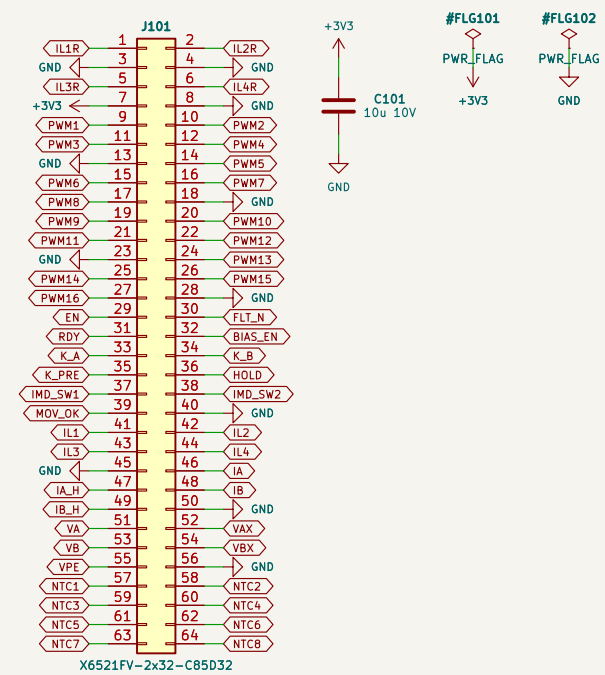
Date: 2026-10-06

Rev: A2

KiCad E.D.A. 10.0.6

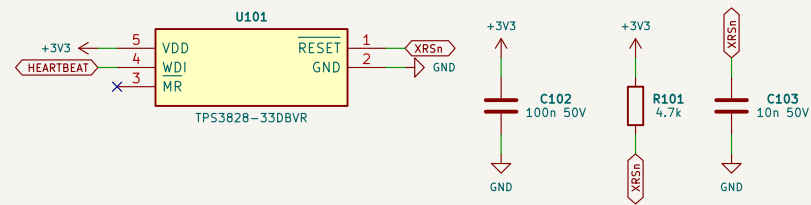
Id: 1/9

PC CONNECTOR (GEN/INTERFACES.PY PC; AGND = GND ON THIS BOARD)



LIVE: GND = AGND = BUS-; nothing on this connector leaves the enclosure. Only +3V3 comes across (the former +24V / +5V pins carry the sensor references IL1R-IL4R since PV-PWR rev A2). +3V3 feeds the MCU, so the power board's 3.3 V signals and VDDIO share one rail and one power-up order (no back-powering).

RESET AND WATCHDOG (TPS3828-33)



RESET (open drain) holds XRSn below 2.93 V and for 200 ms after; no WDI edge for 0.9-2.5 s -> reset. XRSn 4.7k pull-up and 10 nF (SPRSP61C table 5-1: 2.2-10k, <= 100 nF). XRSn also clears the heartbeat monoflop, so every MCU reset sets the trip latch. +5V / +24V supervision: on the power board (RDY).

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 2
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /01 PC connector, supply, reset/
File: 01_supply.kicad_sch

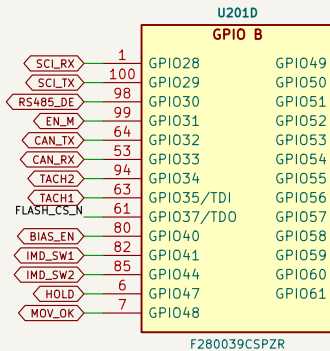
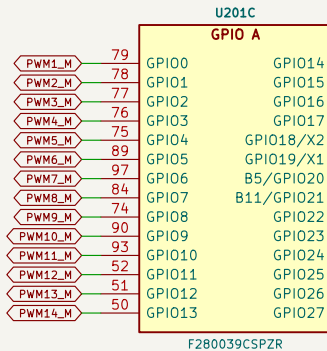
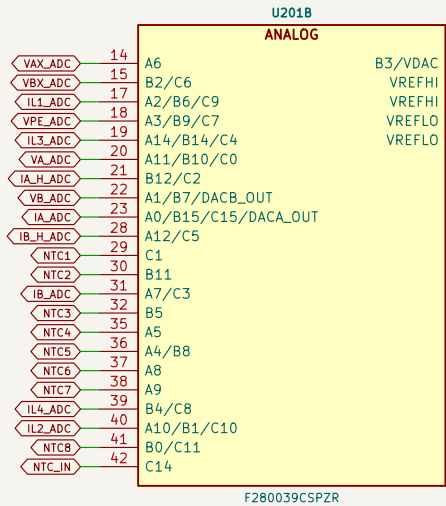
Title: PV-CTL control board - PC connector, supply, reset

Size: A3	Date: 2026-10-06	Rev: A2
KiCad E.D.A. 10.0.6		Id: 2/9

U201A
SUPPLY

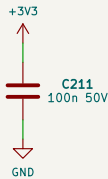
Pin	Function	Value
1	XRS5	2
2	VDD12	3
3	VDD1	4
4	VDD12	5
5	VDD1	6
6	VDD12	7
7	VDD1	8
8	VDD12	9
9	VDD1	10
10	VDD12	11
11	VDD1	12
12	VDD12	13
13	VDD1	14
14	VDD12	15
15	VDD1	16
16	VDD12	17

F28039C50P2R

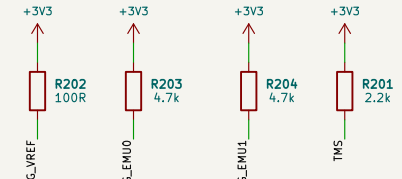
[illegible]

VDDIO 4 x 100 nF ($\geq 0.1 \mu\text{F}$ per pin); VDDA 2.2 μF + 100 nF behind a 300 R bead (pi filter allowed by 6.12.1.4.1); VDD 2 x 10 μF + 2 x 100 nF = 20.2 μF (10–26.8 μF total, internal VREG, 10 % parts).

Pins as gen/data/pv_ctrl_pin_plan.csv (checked against SPRSP61C table 5-1 on every build). VREGENZ = GND; internal 1.2 V VREG; VDD only decoupled.



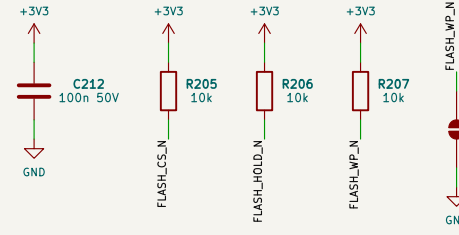
SG-210STF ± 50 ppm (CAN bit timing: CAN boot needs the external clock).
f(X1) 10-25 MHz, tr/tf ≤ 6 ns (SPRSP61C 6.12.3.2.1).



Label: 'HAZARDOUS VOLTAGE - up to 1000 V DC to earth - isolated debug probe only'. cJTAG (IEEE 1149.7, TMS + TCK; SPRSP61C 6.12.6): TDI (3) / TDO (7) open, GPIO35 / 37 are TACH1 / flash CS#. TMS 2.2k pull-up; TRST (2) open; EMU0/1 4.7k; RESET (15) on XRSn. Field update over Ethernet / RS-485 / CAN: no live access.

Pin 1-5 connections for U203:

- Pin 8: +3V3
- Pin 1: FLASH_CS_N
- Pin 6: SPL_CLK
- Pin 5: SPL_SIMO
- Pin 4: GND



GD25Q32E 4 MB on SPIA (SCLK GPIO18, SI GPIO54, SO GPIO55, CS# GPIO37). CS# 10k pull-up: deselected in reset and boot (TDO floats until firmware owns it). HOLD# 10k up. WP# 10k up + write-protect strap JP to GND (closed at end of line with SRP0 = 1: status register and the golden-image blocks locked). Replaces the I2C EEPROM.

Flash boot = GPIO24 high and GPIO32 high at reset (SPRSP61C table 7-8): 10k pull-ups on LATCH_CLR (GPIO24) and CAN_TX (GPIO32, also recessive while the MCU is in reset).

Every MCU output that ends in an AND gate here is low while the MCU is in reset or unprogrammed (GPIOs high impedance): 16_PWM, EN, K_A, K_B, K_PRE, STATUS, FAN_PWM, HEARTBEAT (23 lines, 1 spare element). BIAS_EN, IMD_SW1/2 go straight to PC: pulled down on the power board (10k / 100k, PV-PWR).

Green, lit while the heartbeat monoflop holds WD_OK high (control ISR running, no MCU reset): the hardware's own view of 'firmware alive'. 680 R, ~2 mA from the 74LVC1G123 output (IOH 24 mA at 3.0 V).

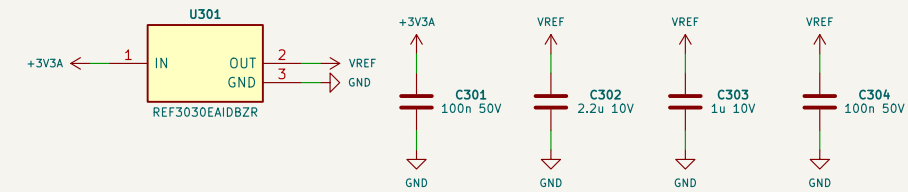
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 3
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /02 F280039C, clock, debug, recorder flash,
File: 02_mcu.kicad_sch

Title: PV-CTL control board – F280039C, clock, debug, recorder flash

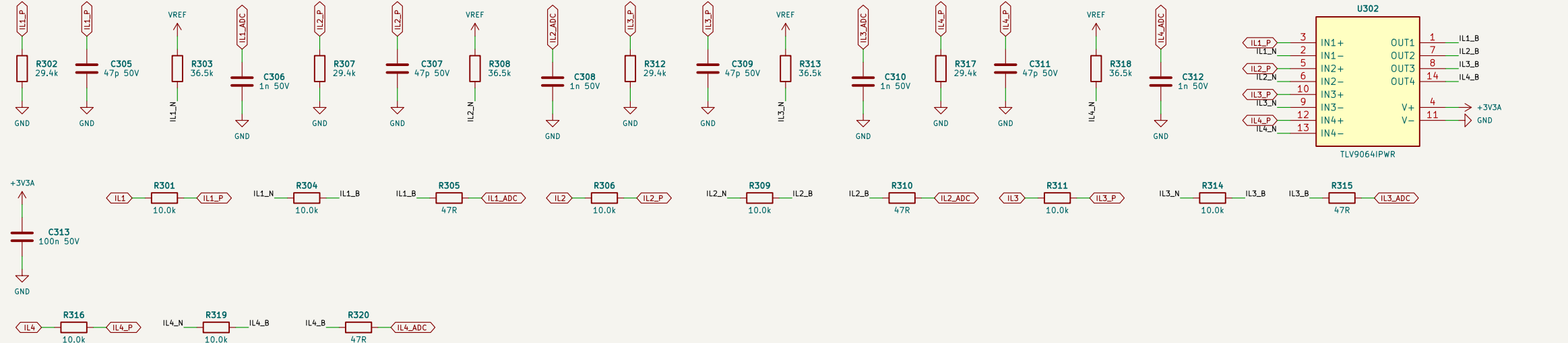
Size: A2	Date: 2026-10-06	Rev: A2
KiCad EDA 10.0.6		Id: 3/9

3.0 V REFERENCE (REF3030E FROM +3V3A)



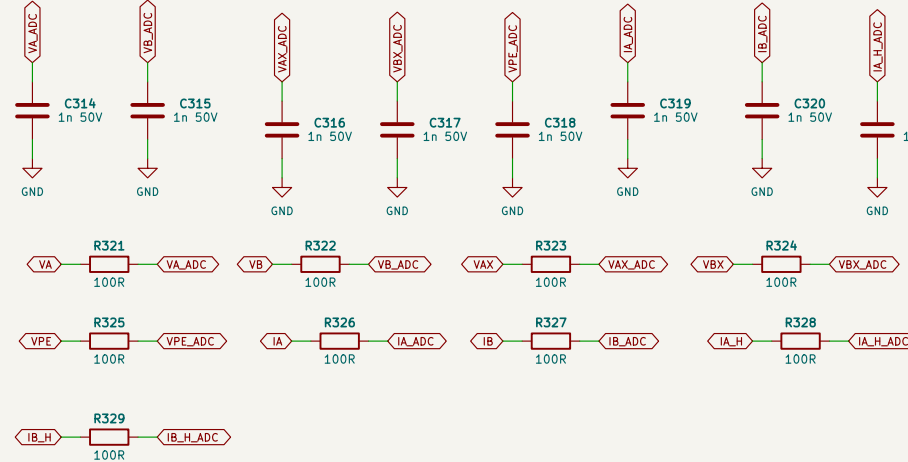
Feeds VREFHI (pins 24/25, >= 2.2 uF), VDAC (pin 16, >= 1 uF), the NTC biases, the IL offset and the threshold ladders of sheet 04; ADC, CMPSS and hardware trips are ratiometric to one reference. It runs from VDDA, so it never sits above VDDA (SPRSP61C 6.12.1.4.2). Needs +3V3 >= 3.20 V (REF30E dropout).

INDUCTOR-CURRENT FRONT ENDS IL1-IL4 (TLV9064 GAIN STAGES)



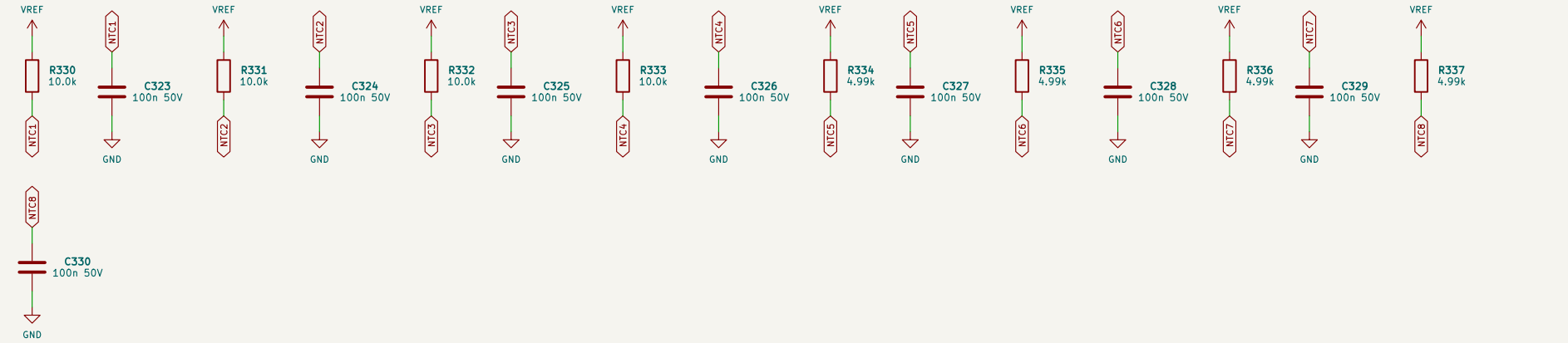
ILn = 2.5 V + 10.667 mV/A (STK-HO/A 75 on PV-PWR, 0.5-4.5 V, not-ratiometric) -> 10.0k / 29.4k 0.1 %
-> ILn_P (47 pF, a = 0.740); trip window (sheet 04) and the + Input: Rg 36.5k to VREF; Rf 10.0k; ADC = 1.555 V + 10.14 mV/A (0.072 A/LSB, -153..+142 A) -> 47 R / 1 nF -> ADC + CMPSS; IL4 = 0 V on PV-P75.

POWER-BOARD ANALOG SIGNALS (VMID 1.641-1.651 V PER PORT, NOT ON PC)



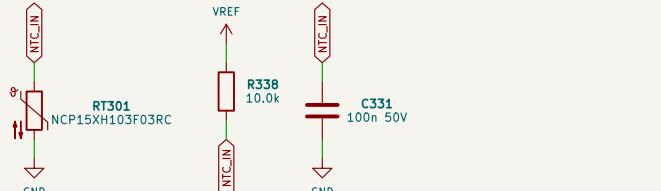
VA/VB/VAX/VBX/VPE = VMID + V/1203; IA/IB = VMID + 3.0 mV/A (+/-478 A); IA_H/IB_H = VMID + 1.0 mV/A. Anti-alias poles are on the power board: here 100 R / 1 nF CGO as the ADC charge bucket. The trip comparators take the PC net itself (no ADC kick-back on them).

NTC INPUTS (10 K B25/100 3988 K TO AGND + 100 nF ON PV-PWR, BIASED HERE FROM VREF)



NTC1-4 heatsink sections (10.0k). NTC5-8 Inductor windings (4.99k), 100 nF each. Open probe = 3.0 V; trips through the open-probe comparators (sheet 04); short = 0 V; trips as over-temperature.

INLET-AIR NTC (ON BOARD)



Murata NCP15XH103F03RC, 10.0k from VREF; ADC only (fan law, derating).

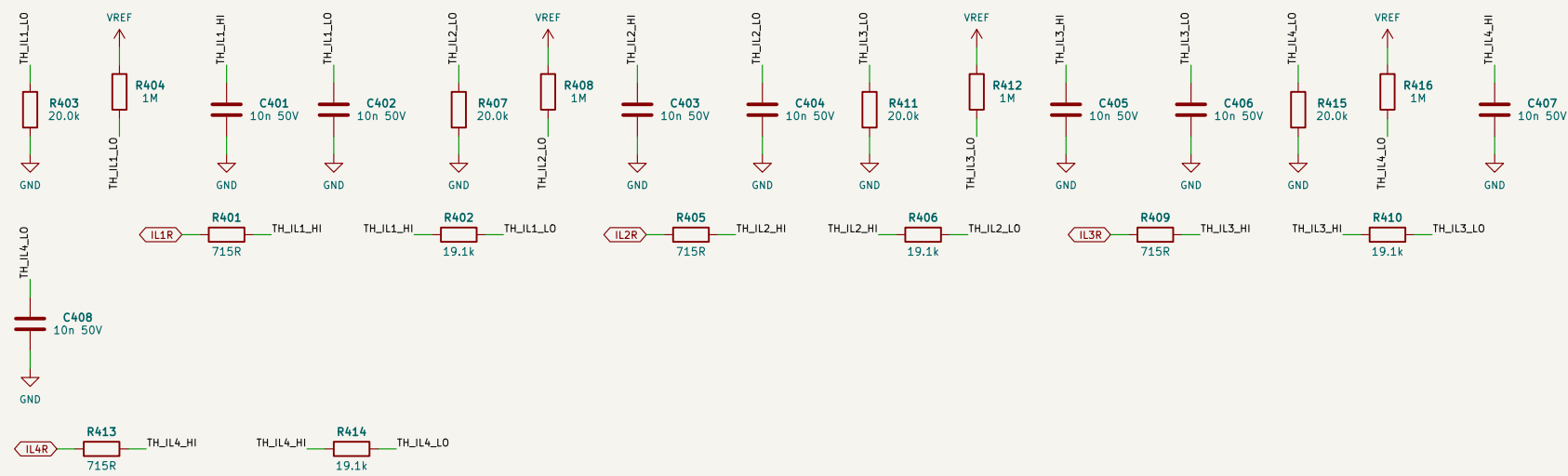
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 4
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /03 Reference and analog front ends/
File: 03_afe.kicad_sch

Title: PV-CTL control board - Reference and analog front ends

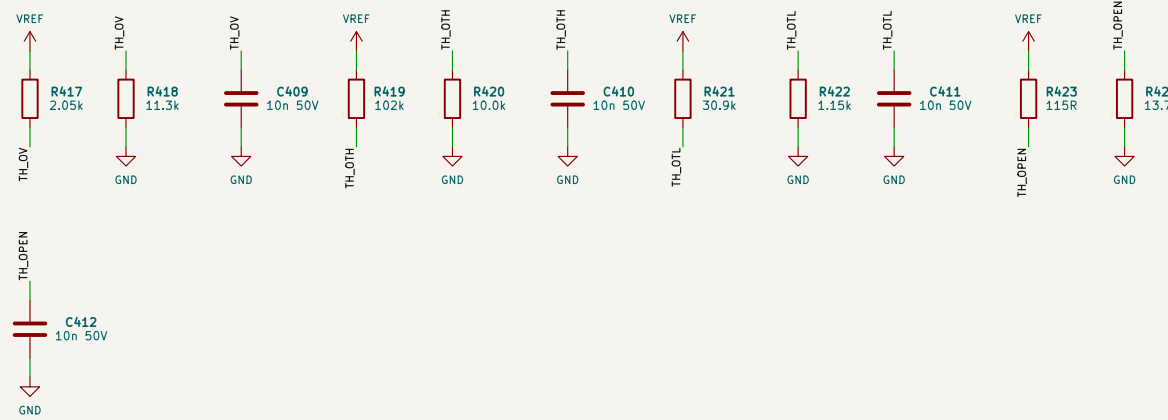
Size: A2 Date: 2026-10-06 Rev: A2
KiCad E.D.A. 10.0.6 Id: 4/9

IL THRESHOLD LADDERS (0.1 %, 25 PPM/K). ONE PER PHASE FROM THAT SENSOR'S UREF (ILNR ON PC)



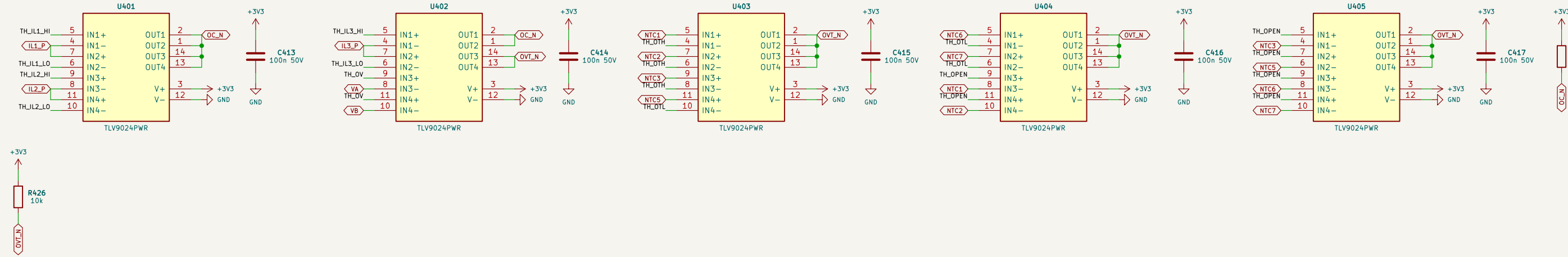
ILNR = 715R - TH_LA_HI = 19.1k - TH_LA_LO = 20.0k - GND; the window follows the sensor's own zero (Vref 2.48-2.52 V is no longer an error term; Vref +/- 0 mV is), 1 M from VREF to TH_LA_LO, with the sensor dead (IL = ILR = 0 V) TH_LA_LO stays -30 mV above ILN_P and the phase trips. Phase 4: fitted on every build.

THRESHOLD LADDERS (0.1 %, 25 PPM/K, FROM VREF)

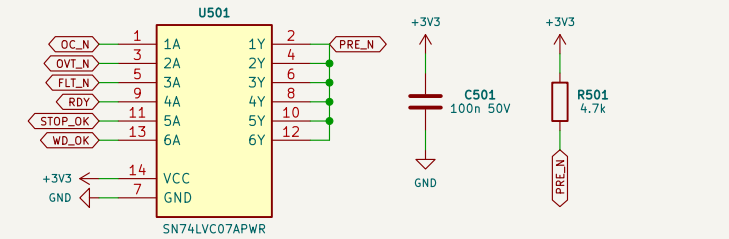


TH_LOV: VA/VB 1075 V; TH_OTH: heatsink 87.9 C (PV-PWR: <= 89.9 C); TH_OTL: Inductor 150 C; TH_OPEN 2.975 V: an NTC node above R is an open probe (cold limit -30 C).

PHASES 1-3 AND THE PORT VOLTAGES (FITTED ON EVERY BUILD)

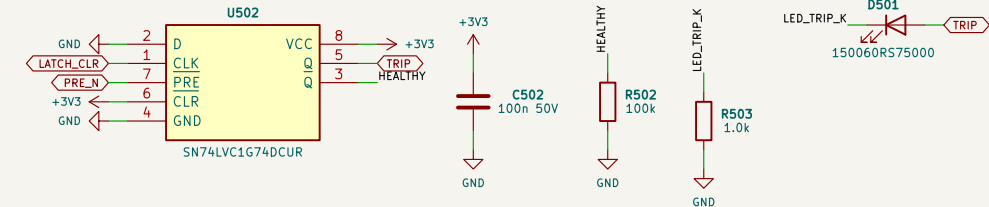


TRIP SOURCES -> PRE_N (OPEN-DRAIN WIRED-AND, 4.7K TO +3V3)



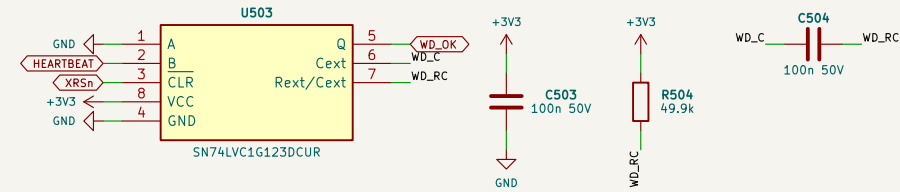
74LVC07 (loff, 5 V tolerant): OC_N, OVT_N, FLT_N (driver DESAT/UVLO + port OC of PV-PWR), RDY (gate supply +5V, live 24 V on PV-PWR), STOP_OK (ENABLE), WD_OK (heartbeat and MCU reset). Any low -> PRE_N low ->

THE LATCH (74LVC1G74, SET-DOMINANT)



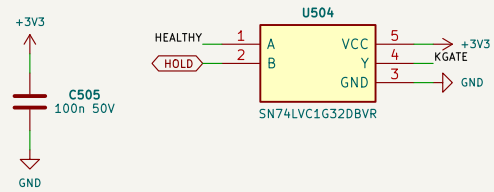
PRE = PRE_N, CLR tied high, D = GND, CLK = LATCH_CLR (GPIO24). A firmware rising edge loads 0 (clears) only while PRE_N is high (no source active); with any source active PRE wins (Table 8-1). Q = TRIP (red LED, INPUTXBAR1 -> T21). /Q = HEALTHY (gates). Unpowered: loff -> HEALTHY reads 0 (100k).

HEARTBEAT WATCHDOG (74LVC1G123)



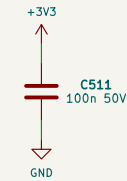
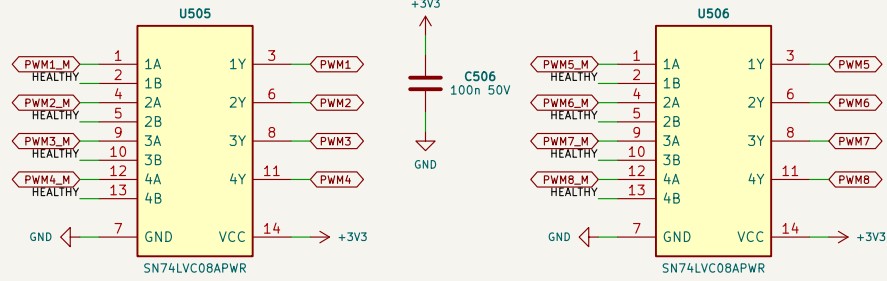
B = HEARTBEAT (rising edges from the control ISR, >= 500 Hz), A = GND, CLR = XRSn, Q = WD_OK stays high while edges come faster than $t_w = 49.9\text{ k} \times 100\text{ n} (4.7\text{--}5.8\text{ ms})$; no edge, a stuck pin or any MCU reset (XRSn low) $\rightarrow$ WD_OK low $\rightarrow$ latch, Rext to VCC, Cext between Cext and Rext/Cext (SC5E586E table 4-1).

CONTACTOR GATE: KGATE = HEALTHY OR HOLD (74LVC1G32)

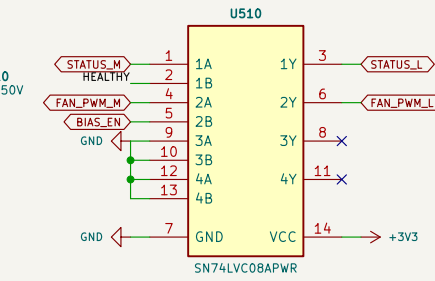
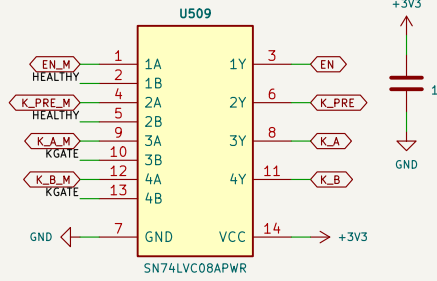
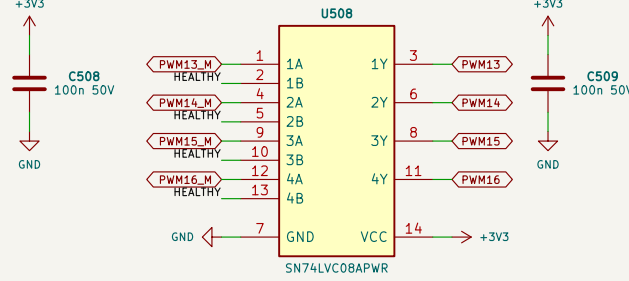
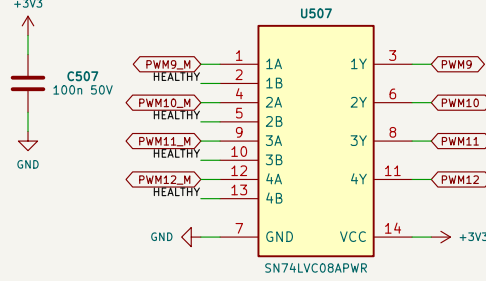


A trip drops K_PRE at once, but K_A / K_B only while HOLD is low: while the (port current above its breaking limit) this board never commands it open; interlock enforces the same. Closing still needs the MCU command AND (HE

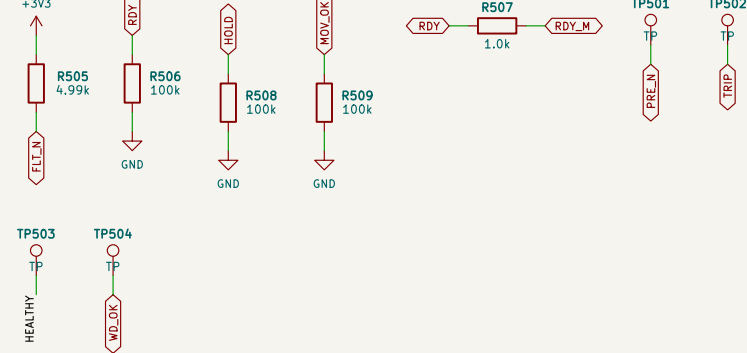
GATING: 6 X 74LVC08 (22 GATES USED, 2 SPARE WITH INPUTS GROUNDED)



PWMn = PWM_M AND HEALTHY (16), EN = EN_M AND HEALTHY, K_PRE = K_PRE_M AND HEALTHY, K_A/K_B = K_A_M AND K_B_M, STATUS_L = STATUS_M AND HEALTHY, FAN_PWM_L = FAN_PWM_M AND BIAS_EN (PV-PWR rule: fans only while the



POWER-BOARD STATUS LINES (PV-PWR AS BUILT)

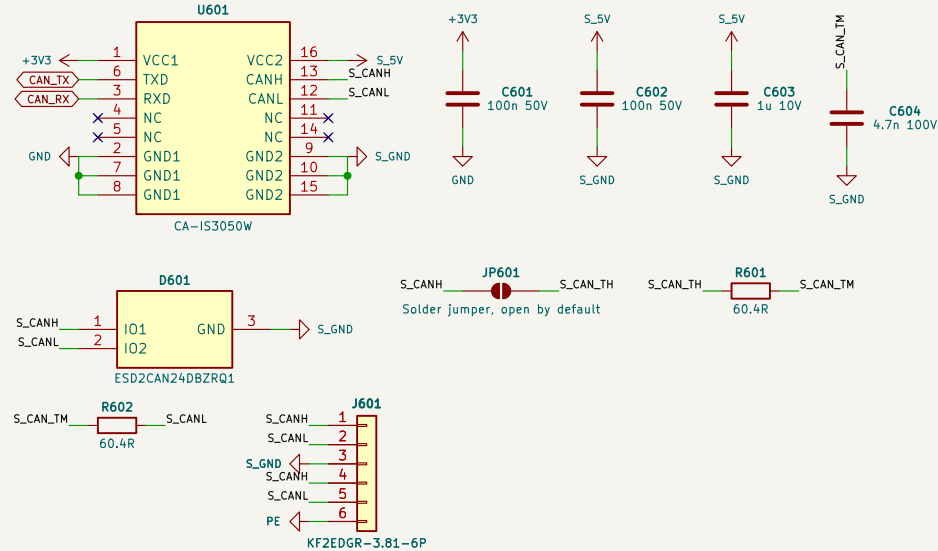


FLT_N: open drain on PV-PWR, pulled up HERE (4.99k, PV-PWR recommendation). RDY: open drain pulled up on PV-PWR (no pull-up here, 100k pull-down allowed), to the MCU via 1k. HOLD push-pull, MOV_OK open drain with 10k on PV-PWR: 100k pull-downs define an unplugged cable (not held / monitor open).

BARRIER B1 RULES (INSULATION_SPEC 'REINFORCED HV-PELV, DC POLE')

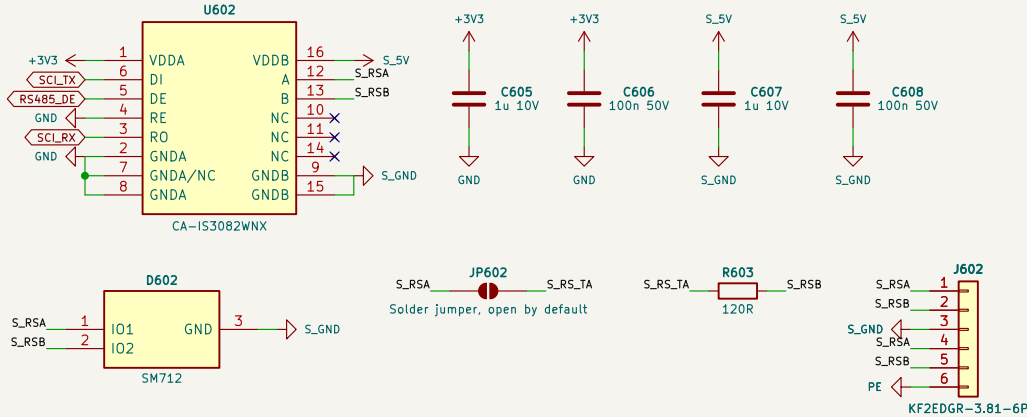
Every part on this sheet that touches LIVE and SELV is a declared isolator: VIOSM >= 8 kV, VISO >= 5 kVrms, VIOVM >= 1414 V DC, CLR/CPG 8 mm, 8 mm < 10 mm (PD2 creepage) -> conformal coating PD1 over the B1 zone is MANDATORY; routed slot under every isolator; SELV copper >= 10 mm from LIVE copper elsewhere.

ISOLATED CAN (BMS / PCS), CA-IS3050W



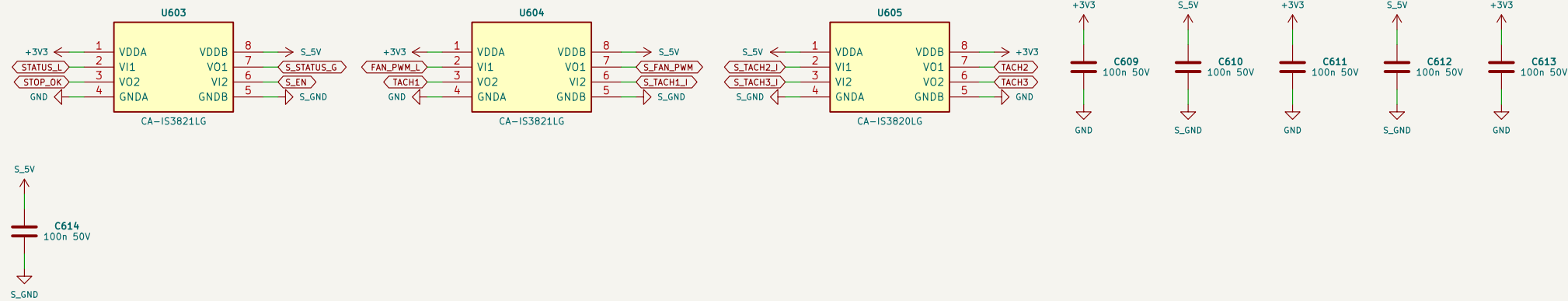
Logic side +3V3 (ICC1 <= 3.6 mA), bus side S_5V (ICC2 <= 73 mA dominant); no live +5V needed, Split termination 2 x 60.4 R + 4.7 nF by JP; ESD2CAN24 to SELV 0 V. Terminal: CANH, CANL, 0 V, CANH, CANL (in / out), shield to PE.

ISOLATED RS-485, CA-IS3082WNX



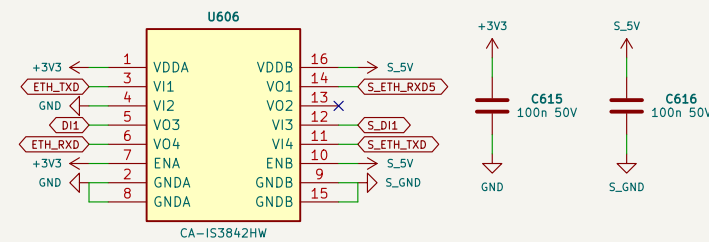
Logic side +3V3; bus side S_5V. DE from the MCU (internal pull-down; driver off in reset), RE tied low (receiver always on, fail-safe), 120 R by JP; SM712 to SELV 0 V. Terminal: A, B, 0 V twice, shield.

DIGITAL ISOLATORS (DEFAULT LOW = SAFE)



3821 #1: STATUS_L -> S_STATUS_G (out), S_EN -> STOP_OK (in; SELV side unpowered or open = STOP).
3821 #2: FAN_PWM_L -> S_FAN_PWM (out; unpowered = fans off), S_TACH1_L -> TACH1. 3820 (A side SELV):
S_TACH2_L, S_TACH3_L -> TACH2, TACH3, 100 nF per side.

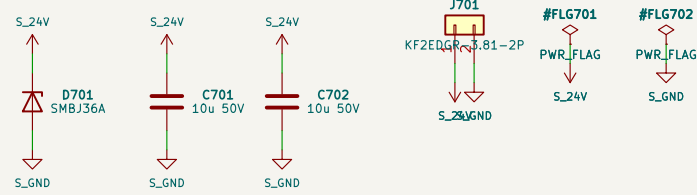
ETHERNET UART AND DI1 ACROSS B1: CA-IS3842HW (REV A2, FITTED ON EVERY VARIANT)



A side LIVE (+3V3): VI1 <- ETH_TXD (SCI8 TX), VO4 -> ETH_RXD (SCI8 RX), VO3 -> DI1 (GPIO52), VI2 spare (GND).
B side SELV (S_5V): VO1 -> 1.0k / 2.0k -> CH9121T RXD1 (3.3 V I/O), VI4 <- CH9121T TXD1, VI3 <- S_DI1 (sheet 07).
Default HIGH; the UART lines idle high with either side unpowered or the bridge not fitted (PV-CTL).

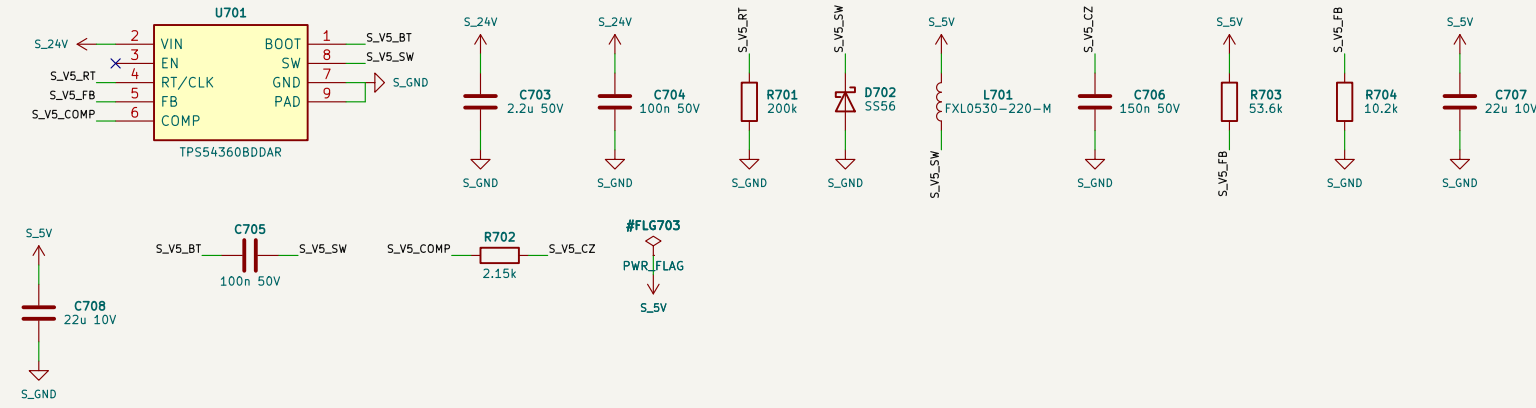
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 7
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

SELV INPUT (2-PIN, DOUBLE-INSULATED LEADS, >= 10 MM FROM LIVE COPPER)



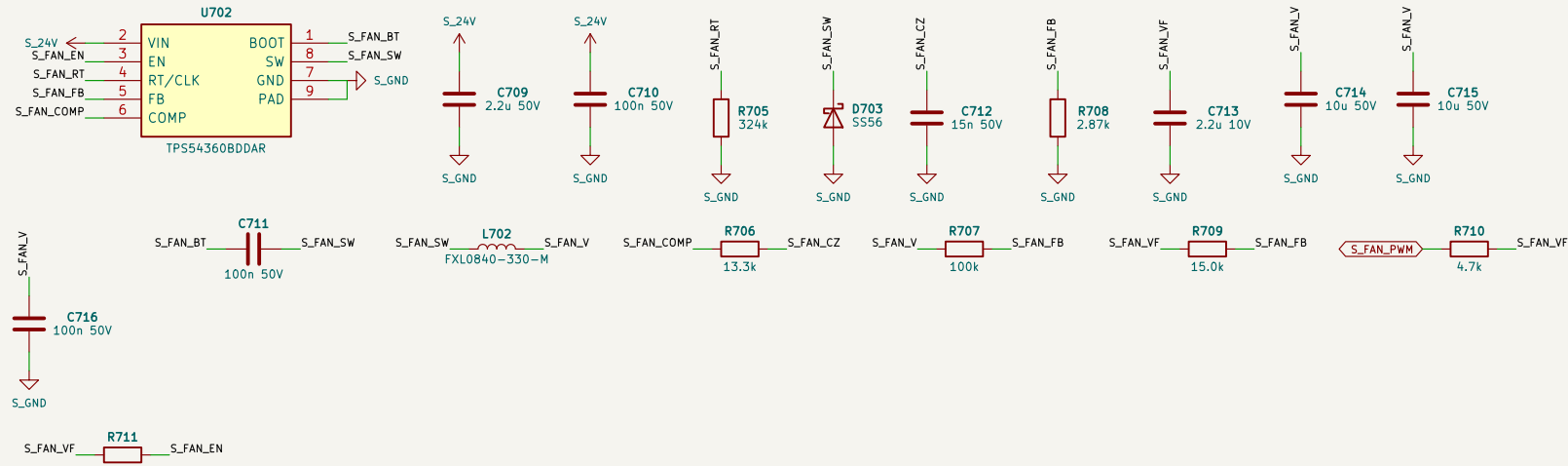
AUX-T1 fan winding, cross-regulated: 12-36 V (sags to -15 V without gate-bias load, PV=PWR), SMBJ36A clamp (36 V working), 2 x 10 uF 50 V. Both bucks have no under-voltage lock-out above 4.5 V.

SELV 5 V (TPS54360B, 500 KHZ)



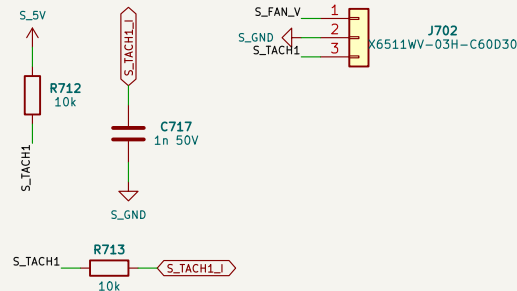
CAN and RS-485 bus sides, SELV side of the three digital isolators: EN floating = on (internal pull-up), 22 uH, 2 x 22 uF.

FAN SUPPLY (TPS54360B, 300 KHZ): SPEED BY VOLTAGE, PASSES WHAT THE WINDING GIVES



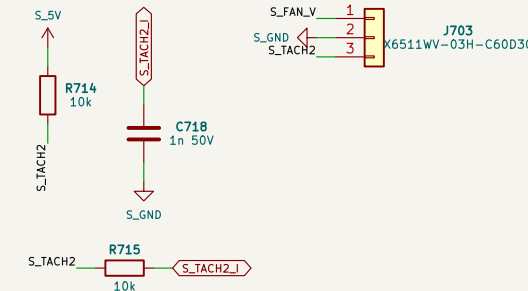
S_FAN_PWM (isolated, 0/5 V) -> 4.7k / 2.2 uF -> S_FAN_VF: EN via 10k (off below 1.1-1.3 V: duty below -25 %, static low at a dead line side = fans OFF) and current injection into FB through 150k: higher duty = lower voltage (24 V ... 7 V): at a low input it runs in dropout, Rtop 100k / Rbot 2.87k.

FAN 1



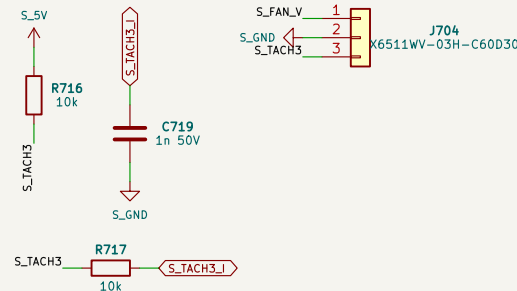
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

FAN 2



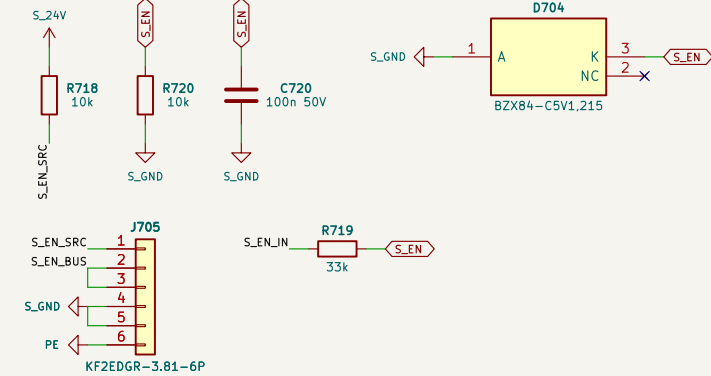
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

FAN 3



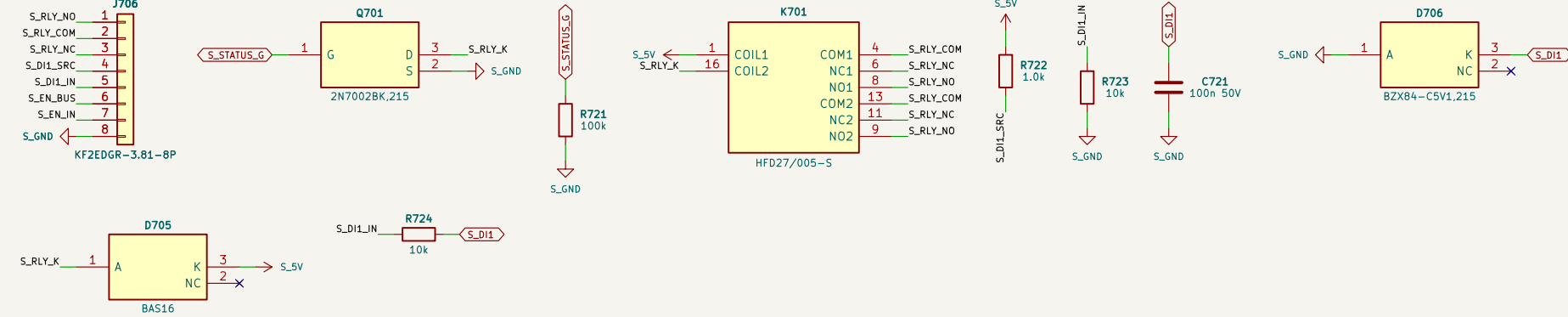
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

STOP INPUT (HARDWARE, CHAINABLE) AND READY PERMISSIVE



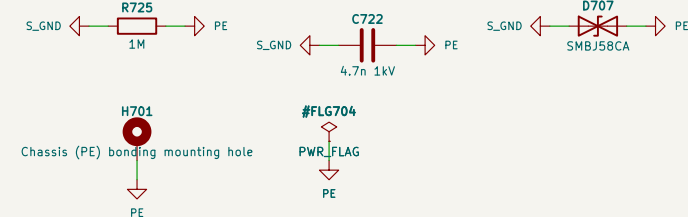
J_STOP 1 S_EN_SRC (S_24V via 10k: dry-contact supply for ONE module), 2 / 3 stop line in / out (loop-through: chain a cabinet 24 V stop line module to module), 4 / 5 0 V in / out, 6 screen to PE, J_ID 6-7: ready permissive in series (wire link if unused) -> S_EN_LN -> 33k / 10k, 5.1 V clamp, 100 nF -> isolator -> STOP_OK -> latch (no firmware): open stop contact, open permissive, broken wire or lost SELV = gates off.

RELAY OUTPUT (READY / FAULT) AND DI1 (BATTERY FAULT / BMS CONTACT)



STATUS channel -> 2N7002BK -> HFD27/005-S coil from S_5V, BAS16 across its: energised = healthy (NO closed): a trip, stop, lost SELV or live side = NC closed. Both poles in parallel: SELV circuits only (<= 30 V AC / 60 V DC), J_ID 4 = 5 V wetting (1.0k from S_5V), 5 = DI1 (10k down, 10k + 100 nF = 1 ms, 5.1 V clamp) -> CA-153842W v13.

SELV 0 V TO PE (SELV MAY BE EARTHED BY THE INSTALLER: PELV)



1 M || 4.7 nF 1 kV plus SMBJ58CA a common-mode surge on the field cables is clamped to PE: a solid PELV bond by the installer is harmless. Plated MS hole = PE / shield bond near the terminals.

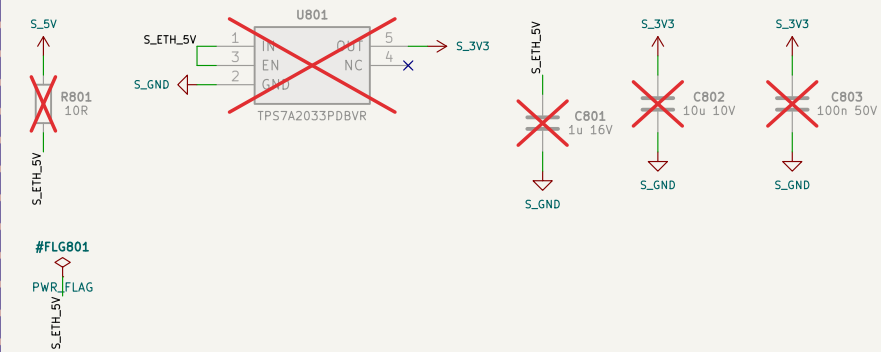
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 8
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /07 SELV supply, fans, stop, DI1, relay/
File: 07_selv.kicad_sch

Title: PV-CTL control board - SELV supply, fans, stop, DI1, relay

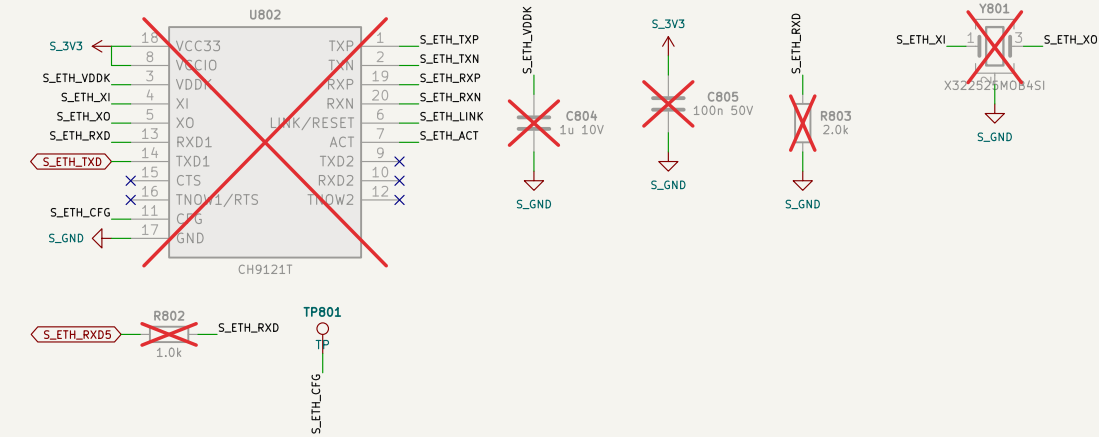
Size: A2 Date: 2026-10-06 Rev: A2
KiCad E.D.A. 10.0.6 Id: 8/9

SELV 3.3 V: TPS7A2033 FROM S_5V THROUGH 10R



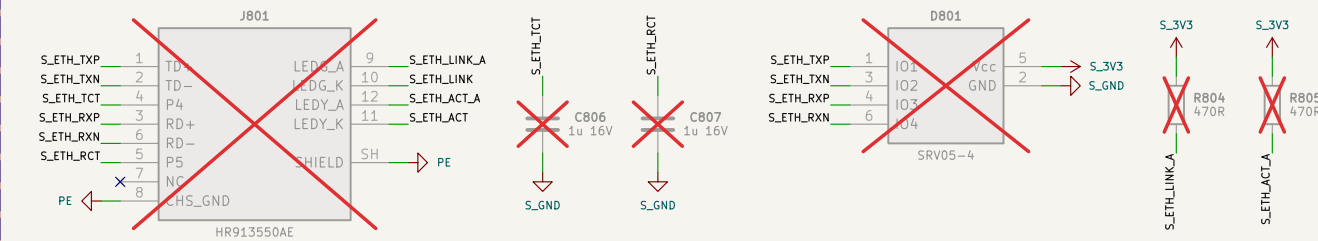
VCC33 needs 3.2–3.4 V (CH9121 V2.6 table 7–2): $\pm 1.5\%$ LDO. The 10R 1206 takes -1 V of the drop so the LDO stays below 125 C at an 85 C board (design check). EN tied to IN. 1 μF in, 10 μF + 100 nF out.

CH9121T (CH9121 V2.6 FIGURE 9–1)



Port 1 (RXD1 / TXD1) to SCIB through the CA-IS3842HW (sheet 06); port 2, CTS, TNOW open. 25 MHz 12 pF crystal, no load C (inside): VDDK 1 uF; power-on reset inside. Straps (pull-ups inside): CFG low at power-up = serial set-up (end-of-line test point); LINK/RESET low for 2 s = factory reset - its LED path holds it high. UART 1.8k / 2.0k.

RJ45 WITH MAGNETICS (HR913550AE), MDI PROTECTION, LINK / ACTIVITY LEDS



Chip side straight to the CH9121T (50 R inside: no 49.9 R); centre taps P4 / P5 to S_GND through 1 uF (maker: no supply on them). Bob-Smith 4 x 75 R + 1000 pF / 2 kV inside, CHS GND and the shield legs to PE (as the bus terminals' screen pins). SRV05-4 on the chip side (0.8 pF). LEDs: green = link, yellow = activity, 470 R.